

1 Periodic functions: Fourier Series

Let $f(t)$ be a periodic function of time with period T :

$$f(t) = f(t+T)$$

The function can be expanded in a Fourier series:

$$f(t) = \frac{1}{2}a_0 + \sum_{k=1}^{\infty} [a_k \cos(k\Omega_0 t) + b_k \sin(k\Omega_0 t)] \quad (1)$$

with:

$$\Omega_0 = \frac{2\pi}{T} \quad (2)$$

and:

$$\begin{aligned} a_k &= \frac{2}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} f(t) \cos(k\Omega_0 t) dt \quad k = 0, 1, 2, \dots \\ b_k &= \frac{2}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} f(t) \sin(k\Omega_0 t) dt \quad k = 1, 2, \dots \end{aligned} \quad (3)$$

Eq (1) is said to be the *trigonometric form* of the Fourier series. Recalling the exponential form of the cosine / sine functions:

$$\cos(k\Omega_0 t) = \frac{e^{ik\Omega_0 t} + e^{-ik\Omega_0 t}}{2} \quad ; \quad \sin(k\Omega_0 t) = \frac{e^{ik\Omega_0 t} - e^{-ik\Omega_0 t}}{2i} \quad (4)$$

it is possible to rewrite the Fourier series expansion of the function as:

$$f(t) = \frac{1}{2}a_0 + \sum_{k=1}^{\infty} \left[\frac{(a_k - ib_k)}{2} e^{ik\Omega_0 t} + \frac{(a_k + ib_k)}{2} e^{-ik\Omega_0 t} \right] \quad (5)$$

Introducing the notation:

$$\begin{aligned} c_0 &= \frac{1}{2}a_0 \\ c_k &= \frac{(a_k - ib_k)}{2} \quad k = 1, 2, \dots \\ c_{-k} &= \frac{(a_k + ib_k)}{2} \quad k = 1, 2, \dots \end{aligned}$$

Eq. (5) becomes:

$$f(t) = \sum_{k=-\infty}^{\infty} c_k e^{ik\Omega_0 t} \quad (6)$$

with:

$$c_k = \frac{1}{T} \int_0^T f(t) e^{-ik\Omega_0 t} dt \quad k = 0, \pm 1, \pm 2, \dots \quad (7)$$

Eq. (6) is known as the *exponential form* of the Fourier series.

Noting that c_{-k} and c_k are complex conjugates, a third from of the Fourier series is obtained as follows:

$$f(t) = F_0 + \sum_1^{\infty} F_k \cos(k\Omega_0 t + \varphi_k) \quad (8)$$

with:

$$\begin{aligned} F_0 &= c_0 = \frac{a_0}{2} \\ F_k &= 2|c_k| = \sqrt{a_k^2 + b_k^2} & k=1, 2, \dots \\ \varphi_k &= \angle c_k = \tan^{-1}\left(\frac{-b_k}{a_k}\right) & k=1, 2, \dots \end{aligned} \quad (9)$$

Considering $f(t)$ as a force representing a periodic excitation applied to a linear, single-input single-output (SISO) mechanical system, the stationary response $x(t)$ of the system can be computed from the frequency response function $G(i\Omega)$ of the SISO system as derived in lesson L02 - *to the excitation Solution of the equations of motion for linear 1 d.o.f. and multi-d.o.f. systems* as:

$$x(t) = X_0 + \sum_1^{\infty} X_k \cos(k\Omega_0 t + \psi_k) \quad (10)$$

with:

$$\begin{aligned} X_0 &= G(i0) F_0 \\ X_k &= |G(i\Omega_k)| F_k & ; & \quad \psi_k = \angle G(i\Omega_k) + \phi_k & ; & \quad \Omega_k = k\Omega_0 \end{aligned} \quad (11)$$

or, equivalently, as:

$$x(t) = \sum_{-\infty}^{\infty} x_k e^{ik\Omega_0 t} \quad (12)$$

with:

$$x_k = G(i\Omega_k) c_k \quad (13)$$

2 Non-periodic functions: Fourier transform

We now consider a non-periodic function of time $f(t)$. The frequency contents of the non-periodic function are given by the *Fourier transform* $F(\Omega)$ of $f(t)$:

$$F(\Omega) = \int_{-\infty}^{+\infty} f(t) e^{-i\Omega t} dt \quad (14)$$

The Fourier transform provides a continuous frequency spectrum, representing the harmonic contents of the non-periodic function, whereas in Section 1 we showed that a periodic function is characterised by a discrete spectrum.

Given the Fourier transform of the signal, the signal in time domain is obtained from the inverse Fourier transform:

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} F(\Omega) e^{i\Omega t} dt \quad (15)$$

It is shown in the book by Meirovitch¹ that the Fourier transform / inverse transforms can be obtained from the exponential form of the Fourier series, Eq. (6), by letting the period T tend to infinity, so that the function of time expressed by Eq. (6) becomes non-periodic and the distance between the discrete frequencies of the discrete spectrum from the series tends to zero, producing a continuous spectrum.

If the non-periodic function of time $f(t)$ represents a time-dependent excitation applied to a linear SISO system, the response $x(t)$ can be obtained as:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(\Omega) e^{i\Omega t} d\Omega \quad (16)$$

with:

$$X(\Omega) = G(i\Omega) F(\Omega) \quad (17)$$

The process defined by Eq.s (16) and (17) to define the response of the SISO system to the non-periodic excitation is the counterpart of the process that can be followed to derive the response of the system to a periodic excitation, defined by Eq.s (12) and (13).

3 Discrete function of time: Discrete Fourier Transform (DFT)

The direct use of the Fourier transform to define the response of a dynamic system to a non-periodic excitation is only possible when an analytical expression is available for both the Fourier transform of the input force, see Eq. (14) and for the frequency response function $G(i\Omega)$. This is not true, however, in most practical cases and therefore an approximation resorting to a discrete version of the Fourier transform can be used instead.

To introduce the discrete approximation of the Fourier transform, known as DFT / Discrete Fourier Transform, we consider the input force is known from a discrete number N of values, $f_0, f_1, \dots, f_{N-1}$ sampled at equi-spaced times $0, dt, 2dt, \dots$. From the N samples of the input force in the time domain, a discrete spectral representation is provided by:

$$\bar{F}_k = \sum_{j=0}^{N-1} f_j e^{-i \frac{2\pi jk}{N}} \quad k = 0, 1, \dots, N-1 \quad (18)$$

The inverse transform of Eq. (17) is provided by:

$$f_j = \frac{1}{N} \sum_{k=0}^{N-1} \bar{F}_k e^{i \frac{2\pi jk}{N}} \quad j = 0, 1, \dots, N-1 \quad (19)$$

We note that for positive values of k higher than $N-1$ and negative values of k Eq. (18) repeats periodically the same sequence of values obtained for $k=0, 1, \dots, N-1$. Furthermore, we note that for positive values of j higher than $N-1$ and negative values of j Eq. (19) repeats periodically the sequence of N sampled values of the function $f(t)$. Hence, the use of a discrete set of data to describe the time-dependent excitation ultimately results in considering the excitation as periodic with period $T=Ndt$. We also note that:

$$\bar{F}_{N-k} = \text{conj}(\bar{F}_k) \quad k = 1, \dots, \frac{N}{2}-1 \quad (20)$$

¹ Meirovitch L.: Fundamentals of Vibrations, Mc Graw-Hill International Edition

A direct relationship can be established between the coefficients of the DFT obtained from Eq. (18) and the coefficients of the Fourier series in exponential form expressed by Eq. (7). Indeed, assuming the input function $f(t)$ is known only from the discrete set of samples $f_0, f_1, \dots, f_{N-1}$, the coefficients of the Fourier series in exponential form for $k = 0, 1, \dots, \frac{N}{2}-1$ can be obtained by approximating the integrals defined by Eq. (7) as:

$$c_k = \frac{1}{T} \int_0^T f(t) e^{ik\Omega_0 t} dt \cong \frac{1}{T} \sum_{j=0}^{N-1} f(t_j) e^{ik\Omega_0 t_j} dt = \frac{1}{Ndt} \sum_{j=0}^{N-1} f_j e^{ik \frac{2\pi}{Ndt} jdt} = \frac{1}{N} \sum_{j=0}^{N-1} f_j e^{i \frac{2\pi}{N} kj} = \frac{1}{N} \bar{F}_k \quad (21)$$

and, using Eq. (8) the coefficients of the Fourier series defined in the form of Eq. (8) for $k = 0, 1, \dots, \frac{N}{2}-1$ are:

$$\begin{aligned} F_0 &= c_0 = \frac{1}{N} \bar{F}_0 \\ F_k &= 2|c_k| = \frac{2}{N} |\bar{F}_k| \quad k = 1, 2, \dots \\ \varphi_k &= \angle c_k = \angle \bar{F}_k \quad k = 1, 2, \dots \end{aligned} \quad (22)$$

Higher harmonics in the Fourier series cannot be obtained due to the sampling frequency used to extract from the continuous time history $f(t)$ the discrete set of samples $f_0, f_1, \dots, f_{N-1}$.

Considering that:

$$\Omega_{\frac{N}{2}} = 2\pi \frac{N}{2T} = 2\pi \frac{N}{2Ndt} = 2\pi \frac{1}{2dt} = 2\pi \frac{f_s}{2} \quad f_s = \frac{1}{dt} \text{ the sampling frequency} \quad (23)$$

we realise that this is consistent with Shannon's theorem about sampling of analog signals (analog to digital conversion /ADC).

The response of the SISO system to a non-periodic excitation defined by a discrete set of N sampled values, $f_0, f_1, \dots, f_{N-1}$ at equi-spaced times $t=j dt$ ($j=0, 1, N-1$), can be obtained in the form of a discrete set of values $x_0, x_1, \dots, x_{N-1}$ of the output sampled at the same times, according to the following equations:

$$x_j = \frac{1}{N} \sum_{k=0}^{N-1} \bar{X}_k e^{i \frac{2\pi}{N} jk} \quad j = 0, 1, \dots, N-1 \quad (24)$$

with:

$$\begin{aligned} \bar{X}_0 &= G(i0) \bar{F}_0 \\ \bar{X}_k &= G(i\Omega_k) \bar{F}_k \quad \bar{X}_{N-k} = \text{conj}(\bar{X}_k) \quad \Omega_k = k \frac{2\pi}{T} = k \frac{2\pi}{Ndt} \quad k = 1, \dots, \frac{N}{2}-1 \end{aligned} \quad (25)$$

Alternatively, the response of the system at the same equi-spaced time values can be obtained according to Eq.s (10) and (11), considering a 'fictitious' periodic excitation defined by a Fourier series in the form of Eq. (8), with coefficients given by Eq. (22).

In both cases, the sequence of the x_j values should not be extended beyond time $(N-1)dt$, as this would result in a periodic time history of the output, which is not consistent with the type of excitation considered.